Health policy 156 (2025) 105316

Contents lists available at ScienceDirect

Health policy

journal homepage: www.elsevier.com/locate/healthpol

Common institutional ownership and the erosion of competition in the American health insurance market: A quantitative analysis

Kevin Smith *

Assistant Professor of Economics and Business at Beloit College, USA

ARTICLE INFO

Keywords:
Common institutional ownership
Affordable care act
Health insurance markets
Private equity

ABSTRACT

Background: Empirical studies have shown that when large institutional investors and private equity funds own a significant stake in multiple publicly traded firms within an industry, this common ownership may lead to anti-competitive behavior. In 2010, the Affordable Care Act (ACA) was passed. This law created health insurance exchanges that allowed for Americans to purchase health insurance plans directly on a marketplace. The insurers on this marketplace have common institutional owners, that could influence the level of competition in these markets.

Objective: To assess the causal effects of common ownership on competition in the US health insurance market.

Methods: To identify the causal impact of common ownership, I use two empirical strategies: the Arellano-Bond Dynamic Panel Model and Bartik-style Instruments. Data on firm ownership came the paper, "Common Ownership Data: Scraped SEC form 13F filings for 1999-2017" by Matthew Backus, Christopher Conlon, and Michael Sinkinson.

Results: The results show that a one standard deviation increase in common ownership leads to a 7.7 % increase in monthly premiums charged for certain health insurance plans. This finding is robust to different model specifications and suggests that common ownership reduces competition in the health insurance exchanges.

Conclusions: The results from this paper indicate that policymakers and regulators need to formulate policies that limit the amount that institutional investors and private equity funds can own of companies in the health insurance industry in the US. Additionally, policymakers need to make more data on firm ownership available, as the data that is available has quality issues.

Research in Context Box

What is already known about the topic: The literature on the effect that common institutional ownership has on competition in product markets is unsettled. There have been several studies that have found positive effects while other studies have found null effects. The reason that this study is needed is because the health insurance marketplace offers a unique setting to study how institutional ownership affects the prices that insurers charge policyholders, a relationship that has not be thoroughly examined prior to this paper.

What does this study add to the literature? This paper adds to this literature by showing that a one standard deviation increases in my common ownership measure results in an increase in health insurance premiums by 7.7 %. This result has not been found before in the literature on common ownership.

What are the policy implications? Policymakers and regulators need to formulate policies that limit the amount that large institutional investors and private equity funds can own of companies in the health insurance industry in the US. Additionally, policymakers need to make more data on firm ownership available, as the data that is available has significant data quality issues.

1. Background

Health care is one of the most expensive commodities consumed in the United States [1]. Americans have the highest per-capita health care spending in the world, and policy makers have debated for decades why the U.S. health care system is so expensive [2]. Some economists have pointed to the multi-payer structure of the health care system as one

* Corresponding author at: 700 College St, Beloit, WI 53511, USA.
E-mail address: smithks@beloit.edu.

https://doi.org/10.1016/j.healthpol.2025.105316
Received 22 January 2025; Received in revised form 3 April 2025; Accepted 6 April 2025
Available online 13 April 2025
0168-8510/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

reason for the higher costs while others have focused on lack of competition in the health care industry, especially among health insurance companies [3,4]. Six companies make up a substantial share of the U.S. health insurance industry, with United Health Care, Centene, Aetna (CVS Health), Humana, Cigna, and Anthem Blue Cross Blue Shield, accounting for more than 40 % of the market in 2020 [5]. The highly concentrated nature of the health insurance market could result in higher prices for policyholders.

To measure market concentration in an industry, economists typically use the Herfindahl- Hirschman Index (HHI). HHI is calculated from 0 to 10,000, with 0 representing perfect competition and 10,000 representing a monopoly. Since HHI measures the concentration in markets only when there is one owner of each firm, recent research suggests HHI levels might not fully reflect con- centration in some markets ([6]a). This is because HHI does not capture common institutional ownership (CO) and industries with higher levels of CO are less competitive. The theory of CO posits that when an institutional investor owns a partial stake in most publicly traded firms within an industry, it could lead to a reduction in market competition [7]. For example, if a diversified owner holds a portfolio of investment in all six of the major health insurance companies, this common owner may be negatively impacted by aggressive com- petition among the firms in their portfolio, where gains by some firms could be offset by losses by other firms. The primary goal of a common owner is to maximize risk-adjusted portfolio returns, not the returns of any individual company ([6]a).

[Insert Table 1 Here]

Table 1 shows private equity funds and institutional owners of the six largest publicly traded health insurance companies in the US as of the third quarter of 2024 [8]. A key observation from Table 1 is that the Vanguard Group owns at least a 5 % share in all of the major health insurance providers, while Blackrock, T.Rowe Price, and FMR Corp also have significant investments in almost all the insurers. These institutional investors and private equity firms are clearly common owners as they have invested in most of the largest companies in the health insurance industry.

A key area of investigation in the CO literature is the mechanisms that a common owner might use to directly or indirectly influence managers of the companies in their portfolio. Azar et al. [6] found that in the airline industry there was a direct channel that common owners used to influence managers, through changes in corporate governance policies. They also identified indirect channels where common owners could give managers a "quiet life" by not pressing them to compete more aggressively with rival firms, thereby reducing competition. Anton et al. (2018) also found alternative channels through which CO directly affects firms. One such channel is through the incentives from common owners for managers to invest more in Research and Development (R&D). This is due to the fact that common owners see spending on R&D as a predictor of future profitability, and that R&D investments usually do not cause greater competition between rival firms (Anton et al., 2018).

The goal of this paper is to test whether increases in CO could lead to increases in health insurance premiums. In competitive markets, the ownership structure of a firm or multiple firms in an industry should not affect prices. This is because prices in these markets should be primarily driven by changes in the micro and macro-economic conditions in which they operate. So, changes in health insurance premiums can be used as a proxy for the overall competitiveness of the market because in competitive markets, common owners should not be able to affect product prices. In order to measure CO, this paper uses a Modified Herfindahl-Hirschman Index (MHHI). MHHI was developed to measure the effects of cross-stock ownership by competitors in the same industry [7]. The benefit of using MHHI is that it can be deconstructed into two components, namely HHI, which measures the relative competitiveness of the industry and MHHI delta (MHHID) which measures the density of share ownership of competitors by a common owner. MHHID can therefore be used to measure the share of network ownership that is held by common owners within an industry.

A major empirical challenge is that MHHID is potentially endogenous to product prices. Institutional investors do not invest in firms at random, so changes in CO could be directly related to expected changes in profitability of the industry or certain market segments. To address the endogeneity of MHHID I use the Arellano-Bond (ARB) dynamic panel model and Bartik-style instruments. The ARB model uses lags of the dependent and endogenous variables to correct for endogeneity in a first difference regression. In contrast, Bartik-style instruments are based on the interaction of local industry shares and national industry growth rates.

This paper adds to the existing literature in several important ways. It is the first paper, to the best of my knowledge, that analyzes the effect of CO on health insurance premiums. In addition, I use two very distinct identification strategies to account for the endogeneity of MHHID, allowing for a more robust test of the theory of CO relative to the prior literature. Furthermore, I utilize a higher quality data set, than those used in previous CO papers. This data was collected in Backus et al. [9], and fixes issues with firm ownership data that has been criticized extensively in the literature [10].

1.1. Health insurance exchanges

One of the primary provisions of the Patient Protection and Affordable Care Act (ACA) was the creation of health insurance exchanges (HIE) that allow most Americans to purchase private health insurance plans directly from insurance companies and get subsidies for those plans based on their income [11]. HIE plans are not available to Medicaid enrollees. In 2014, these marketplaces opened across the country, with some organized by the federal government and others by state governments. Initially, plans offered on the exchanges were less expensive than what many analysts had expected, and there was fierce competition between insurance companies [12]. However, by 2016, many insurance companies reported major losses, because new customers on the exchanges were sicker than initially estimated [13]. With their losses mounting, many insurance companies threatened to pull out of marketplaces that were unprofitable, and in 2017 many did, resulting in several markets with only one insurance provider [12]. As a result,

Table 1
Top Institutional Investors in the Six Largest US Health Insurers.

Institutional Investor	United Health Care	Humana	Centene	Cigna	CVS Health	Anthem
Vanguard Group Inc	10.33 %	9.69 %	10.6 %	7.88 %	8.15 %	7.34 %
Blackrock Inc	8.94 %	9.90 %	6.24 %	8.45 %	7.31 %	4.65 %
FMR Corp	3.95 %	4.24 %	5.59 %	5.17 %	–	2.38 %
State Street Corp	5.53 %	4.69 %	4.03 %	–	4.13 %	–
T. Rowe Price Associates	3.72 %	4.57 %	–	7.09 %	–	5.1 %
Capital World Investors	–	–	7.68 %	4.42 %	3.49 %	–
Wellington Management Co	–	5.35 %	–	–	–	4.39 %
Capital International Investors	–	–	–	–	2.30 %	–

Note: This table shows the percentage ownership share of the six largest American health insurances companies by institutional investors, in the third quarter of 2024. Source: [8].

many customers saw their premiums for marketplace plans rise by as much as 15 % nationally as insurance companies scrambled to regain lost revenue [13]. In addition, insurance companies started to offer high-deductible plans in the exchanges, which allowed the insurance companies to regain profitability [14].

The restructuring of the health insurance plans offered on the marketplaces did not lead to a return to the high levels of competition evident during the first year of the exchanges [15]. This is in part because the number of plans offered on the exchanges dropped from 2015 to 2018, and some markets became monopolies [12]. Over time, the ACA exchanges stabilized. A 2021 Kaiser Family Foundation (KFF) study found that insurer participation levels steadily climbed, after 2018, to levels seen in the early years of ACA roll out, see Fig. 1 (refer to supplementary material) [16].

An important question is why competition on the exchanges took so long to recover. One explanation is, that between 2017 and 2018 the federal government defunded the cost-sharing reduction subsidies in the ACA. In response, insurers in more than 40 states were forced to increase premiums for their silver plans to keep their plans actuarially fair (this is referred to as silver loading) [17]. The implication of this policy change destabilized the ACA markets resulting in significant premium increases seen during this time period.

Another possible explanation is that the health insurance industry is already highly concentrated with there being only a handful of large health insurance providers in the industry [13]. In 2015, the industry almost consolidated further as Anthem announced it intended to purchase Cigna, and in the same year Aetna attempted to purchase Humana [18]. The Federal Trade Commission (FTC) and Department of Justice (DOJ) citing antitrust concerns, moved to block these acquisitions. After Aetna's failed merger with Humana, Aetna chose to leave the ACA exchanges altogether in 2017, even leaving markets such as Nebraska and Delaware that were profitable ([19]; Luhby, 2016).

1.2. Prior research

Concerns over CO have existed in one form or another going back to the 1980s. Rotemberg [20] posited that if investors owned a slice of every firm in an industry, they would maximize risk- adjusted portfolio value if firms competed less and collectively raised prices. However, due to the lack of empirical evidence, CO was viewed as only a theoretical construct. Azar et al. [6] renewed interest in CO by demonstrating a causal link between CO and product market competition, specifically airline tickets. Azar et al. [6] used the acquisition of Barclays Global Investors (BGI) by the institutional investor Blackrock in 2009 to identify their estimates. Azar et al. [6] treats this shock as exogenous, as Blackrock overnight significantly increased its cross holdings in the airline industry as a by-product of the acquisition of BGI. Azar et al. [6] found that an increase in CO due to the merger led to an increase in the average ticket price paid by airline travelers.

Subsequent studies disputed the findings in Azar et al. [6]. In particular, Dennis et al. [21] showed that the results found in Azar et al. [6] are sensitive to variations in their econometric model. Also, Kennedy et al. [22] using the same data as Azar et al. [6] found no effect of CO on airline ticket prices when using a structural model with a nested logit demand function. However, a subsequent paper by Azar et al. [23] pointed out that the papers that tried to replicate their work were either not looking at the same data or used models that were very dissimilar from the ones they used in the original study.

Schmalz [24] conducts a broad review of the literature surrounding CO and finds that researchers have been able to confirm that increases in CO have broad anti-competitive effects using different structural and reduce-form models across various settings. Schmalz [24] also finds that the effects of CO are heterogeneous and can vary greatly across industries and countries. There are various channels through which CO can incentivize anti-competitive behavior. For instance, Schmalz [24] shows that stricter disclosure requirements for publicly traded firms may weaken competition among commonly owned firms. This is because privately held information has strategic value and revealing this information allows competitors to better coordinate their responses to actions taken by the revealing firm. Such coordination reduces competition in the industry, benefiting common owners.

Additionally, papers such as He and Huang [25] find that having common owners across firms can lead to stronger coordination among firms. Firms that are cross owned experience an increase in their market shares when there is an increase in CO. He and Huang [25] use bank mergers to identify the effect of CO, because mergers result in an exogenous change in ownership shares. Irani et al. [26], also illustrates that if two firms have common owners, it leads to a 45 % decrease in the likelihood of a competing bid in the acquisitions market. Irani et al. [26] uses a similar empirical approach as He and Huang [25], but the authors also create instrumental variables for CO using lagged values of their ownership variable.

A major factor limiting CO research is access to high quality stock ownership and product- market data. Backus et al. [9] attempts to overcome the deficits in share ownership data. The data used in most papers comes from the SEC's 13F holdings accessed through the Thomson- Reuters Spectrum database, which has been known to have some deficiencies, including missing and incorrect data [10]. Instead, Backus et al. [9] scraped information directly from 13F forms. By doing so, they created a data set that is more reliable than the Spectrum database.

Finally, there is a growing strand of the CO literature that has found positive effects of CO. Antón et al. [27] found that firms often face many disincentives to innovate. This is because they may not fully capture the benefits of their inventions before other firms can leverage these innovations to their advantage. However, Antón et al. [27] demonstrate that CO can mitigate this problem and foster greater corporate innovation, by aligning incentives across firms. The authors found that CO can promote innovation in firms with shared technological interests while having nuanced effects depending on the firms' relative use of technology in their production process. This result highlights that the effect of CO on a host of market outcomes could be ambiguous and emphasizes why a paper examining the effect of CO on health insurance premiums is so important and timely.

1.3. Data overview

My data comes from the ACA public use files provided by the Center for Medicare and Medicaid Services (CMS). These data include information on monthly insurance rates (premiums), cost sharing, plan types, plan quality, and services offered in certain plans, such as wellness programs. The CMS's ACA marketplace database contains data on only the 35 states that use the federal government's health insurance exchange, healthcare.gov, from 2014 to 2019. This is due to the fragmented nature of the ACA exchanges, with some states managing their own exchanges while others choosing to allow the federal government to manage their exchange.

CMS reports on health insurance premiums as individual rates. The individual rate is defined as the dollar value for monthly insurance premiums paid by a non-tobacco user. Individual rates on the insurance exchanges are determined by the policyholder's age, tobacco use, and geographic location. I use the individual rate for non-tobacco users as the primary outcome variable in this paper. CMS also provides information on a plan's quality within a tier system, where they assign metals to insurance plans based on the generosity of the plan (platinum, gold, silver, bronze). I subset to silver plans for the empirical analysis in order to reduce heterogeneity in plan generosity. Silver plans are purchased the most frequently by consumers on the exchanges and are the most likely to be offered by insurers across the sample period. Using the silver plan as a benchmark is also common practice in the literature [28].

I also obtained data for several control variables. These include the unemployment rate from the Federal Reserve Bank of St. Louis (FRED),

Gross Domestic Product (GDP) per capita from the Bureau of Economic Analysis, the state obesity rate from the Robert Wood Johnson Foundation, enrollment in all private insurance plans from the Kaiser Family Foundation, and the level of education attained, the poverty rate, median household income, Medicare and Medicaid enrollment from the Census Bureau.

1.4. Common institutional ownership measures

I use quarterly data on share ownership by institutional investors from 2014 to 2019, collected by Backus et al. [9] from the SEC 13F holdings. 13Fs are filed by all institutional investors quarterly who hold over a $100 million in securities under their management [26]. Previous studies have mostly used the Thomson-Reuters Spectrum database to analyze share ownership data and to create their measure of CO. However, there have been significant documented problems with the Spectrum data, including missing and incorrect data found throughout the database.

Additionally, I use this data to create MHHI. Unlike the standard HHI which measures changes in industry concentration from differentiated market share owners, the MHHI includes a measure that considers the ownership structure of firms. O'Brien and Salop [7] first showed that MHHI can be deconstructed to include both HHI and a measure for commonly owned shares as shown in Eqs. 1 and 2.

$$MHHI = \sum_j S_j{}^2 + \sum_j \sum_{k \neq j} S_k S_j \left(\frac{\sum_i \gamma_{ij} \beta_{ik}}{\sum_i \gamma_{ij} \beta_{ij}} \right) \tag{1}$$

$$MHHI = HHI + \sum_j \sum_{k \neq j} S_k S_j \left(\frac{\sum_i \gamma_{ij} \beta_{ik}}{\sum_i \gamma_{ij} \beta_{ij}} \right) \tag{2}$$

j and k index the firms and their competitors in the health insurance industry, i indexes' the institutional owners, S represents the insurers' market share, γ is the fraction of ordinary shares owned by the institutional investor and β is the fraction of control shares owned by institutional investors.

This measure of commonly owned shares is called MHHI delta (MHHID) and measures the extent to which firms within an industry are interconnected by a network of common owners. At higher levels of MHHID there is more CO in the industry. The MHHID can be written as shown in Eq. 3 ([6,7,29]a).

$$MHHID = \sum_j \sum_{k \neq j} S_k S_j \left(\frac{\sum_i \gamma_{ij} \beta_{ik}}{\sum_i \gamma_{ij} \beta_{ij}} \right) \tag{3}$$

The key difference between γ and β is that control shares give holders more voting rights than common shares. Control shareholders have sole or shared voting power, as these shares give them the power to vote or direct the exercise of voting for a security ([6]a).

There are several considerations in the construction of the MHHID. First, not all common owners are observed because some owners have a level of ownership that is too small to be influential. The owners that are observed are those referred to as beneficial owners or blockholders. Blockholders are shareholders that own at least 5 % of the control shares of the company. He and Huang [25], Azar et al. [6], and Irani et al. [26] all found that 5 % is the minimal level of ownership that common owners need to be able to exert significant influence over firms in their portfolios.

Most institutional investors that are observed in the data had at least 5 % ownership in the health insurance companies that provided plans on the ACA exchanges. Secondly, I restrict the CO measure to plans from the six largest health insurance companies, namely United Health Care, Centene, Aetna (CVS Health), Humana, Cigna, and Anthem Blue Cross Blue Shield, which composed more than 40 % of the private health insurance market in 2020 [5]. The vast majority of the plans listed on the exchanges came from these six companies and those that did not, were from smaller insurers. Smaller insurers are often state- specific and in many cases offered plans for only one year. Also, common owners are less likely to be blockholders in these smaller insurers.

1.5. Summary statistics

[Insert Table 2 Here]

Table 2 presents the summary statistics of the variables used in the analysis. The HHI measure is relatively small, with a mean of 1225, and a median of 1031. This is expected given HHI's limitations in capturing CO. In contrast, the MHHID, which accounts for these effects, reflects a higher relative level of concentration, with a mean of 0.135, and a median of 0.107. This underscores the importance of using MHHID to measure market concentration as opposed to just using HHI.

Table 2 also shows that the average monthly individual rate for silver plans is $500, which is consistent with Diaz et al. [30] finding that the average monthly premiums paid by most Americans was $468 in 2019. Health Maintenance Organizations (HMO) make up 51 % of the plans in the sample while Preferred Provider Organizations (PPO) made up 26 %. Average out of pocket spending is $190 during the sample period.

All the variables listed on Table 2 were standardized when included in the regression models. Standardizing all the variables was done to make the interpretation of the variables of interest more precise. When standardized, each variable is interpreted as a one standard deviation change in the explanatory variable. Standardization also puts the variables into the same unit of measurement.

In Fig. A2, Panel A, I present a histogram showing the distribution of individual rates over the sample period. Fig. A2 shows that the individual rate is right-skewed. This skewness is the result of several factors. One factor could be that states, after the messy roll out of the ACA exchanges in 2014, allowed insurers to drastically raise rates in the following years in order to recoup losses that they had suffered on the exchanges. Another factor could be the inclusion of different plan types with premiums that their generosity. 51 % of plans in our sample are HMOs, which are relatively inexpensive compared to PPO plans, which make up 26 % of the sample.

If not corrected, the right skewedness could affect my results, so I transformed the individual rate variable using the logarithmic transformation and present the transformed variable in Fig. A2, Panel B.

2. Methods

One of the biggest challenges in the analysis of CO is the issue of endogeneity or simultaneity bias in Eq. 4. Large institutional investors do not invest in companies or industries at random. In practical terms what may seem like CO affecting insurance premiums could instead simply be institutional investors buying shares of companies or industries that they view as already performing well. To correct this identification issue, the CO literature has used a variety of identification strategies. One popular strategy is the use of mergers and acquisitions of institutional investors in a difference-in-differences (DID) framework. Azar et al. [6] and He and Huang [25] both use the DID strategy to identify their estimates. Unfortunately, during my sample period there were no significant mergers or acquisitions of institutional investors.

Alternatively, Kwon [31] uses the addition of firms to the Standard and Poor (SP) Composite 1500 as an instrumental variable, under the assumption that institutional investors frequently make investments that reflect this index. Participation in the index is therefore correlated with CO, but it is unlikely to be correlated with the unobserved determinants of the outcome variable. Alternatively, Semov [29] exploits an exogenous shock to CO due to sudden capital outflows related to the 2003 mutual fund scandal. He creates a MHHID measure that only contains holdings by mutual funds that were involved in the scandal and then uses the sudden increase in capital outflows from the scandal to predict future changes in MHHID.

Irani et al. [26] and Connelly et al. [32] both use lagged values of

Table 2
Summary Statistics.

Variable	Mean	Median	Std. Dev.	Min.	Max.	N
Business Year	2017	2016	2	2014	2019	7789
State Code	23	25	12	1	35	7789
Individual Rate ($)	499	473	139	200	1196	7789
MHHIdelta	0.135	0.107	0.032	0.103	0.187	7789
HHI	1225	1031	894	283	7532	6107
Ever Enrolled	5108	1014	16,186	0	373,872	5859
Dis Enrolled	936	211	2619	0	45,964	5306
Mean Out-of-pocket Spending	191	192	38	83	332	7721
GDP Per capita	45,944	45,845	6240	31,522	73,478	6166
Obesity rate	0.311	0.313	0.027	0.221	0.395	6166
Poverty Rate (%)	14	14	2.4	8	22	6651
Poverty Rate for Children Under 5 (%)	22	22	4.3	8.7	34.6	6651
Median Household Income ($)	55,633	55,433	6777	39,738	90,395	6651
Total Educational Attainment 18 to 24	945,943	768,272	728,321	51,505	2,826,700	6585
Less than High School	115,312	78,214	116,765	8.5	445,677	6585
High School Graduate	266,335	203,312	249,416	23.3	977,096	6585
Some College	364,458	271,584	332,445	40.1	1,224,498	6585
Bachelors or Higher	82,720	64,616	71,519	5	267,539	6585
Enrollment in Original Medicare	1,131,317	1,048,189	734,287	66,058	2,570,712	6585
Enrollment in Medicare Advantage	561,517	425,454	466,114	22	2,063,774	6585
Exclusive Provider Organization (EPO)	0.133	0	0.34	0	1	7789
Health Maintenance Organizations (HMO)	0.515	1	0.5	0	1	7789
Point of Service (POS)	0.089	0	0.285	0	1	7789
Preferred Provider Organizations (PPO)	0.262	0	0.44	0	1	7789

Note: This table shows the summary statistics for the variables used. Displayed are the mean, the median, the standard deviation, the minimum value, the maximum value, and the number of observations. The sample presented is comprised of data from the 35 states that were run by the federal government between the period 2014-2019. All variables are standardized in the empirical analysis.

their CO measure to identify their empirical model based on the assumption that the level of CO in the past does not affect the pricing decisions made by managers in the current period [33]. This is particularly true with the plans offered on the ACA exchanges. Plans are bought yearly during a specific window of time and plans can vary from year to year. The prior ownership structure of insurers should not affect their current pricing decisions. This strategy has several shortcomings. If the outcome is serially correlated then the estimates will be biased [32].

In this paper, I used several different strategies to address the endogeneity of the CO measure. The paper uses the Fixed Effects Approach (FE), Instrumental Variable Approach (IV), Arellano-Bond Dynamic Panel model (ARB), and Bartik Instruments (Bartik).

2.1. Fixed effects approach

In the fixed effect model, I specify the log of individual rate as a function of MHHID, HHI, a set of state and plan level controls and year and state fixed effects. I present the model in equation 4;

$$\log(Rate_{pjt}) = \beta_0 + \beta_1 MHHID_{jt} + \beta_2 HHI_{jt} + \beta_3 X_{jt} + \alpha_t + \gamma_j + \epsilon_{pjt} \quad (4)$$

$Rate_{jpt}$ represents the insurance premium charged monthly for a benchmark silver plan p, in state j, at time period t. $MHHID_{jt}$ measures CO, HHI_{ij} measures market concentration, X_{jt} includes the control variables, α_t is a year fixed effect, and γ_j is a state fixed effect.

This model follows the work done in Lewellen and Lowry [10], Azar et al. [6], and He and Huang [25]. The key benefit of this model is that it uses variation in rates and MHHID overtime, within states for identification. Thereby removing time invariant state specific unobservables.

2.2. Instrument variable approach

To instrument for the possibly endogenous measure of CO, MHHID, I use two lagged values of MHHID, which are strong instruments. Table 3 shows the first-stage of the two-stage least squares estimator used to implement the instrumental variable approach. The F-statistic is 25.6, which is significantly above the standard used in the literature of 10 [34]. The standard errors in the first stage were clustered at the state level. Furthermore, the Hanson's J test for over identification indicates that I do not reject the null hypothesis that the IVs are valid.

[Insert Table 3 Here]

Table 3
First-Stage Instrument Power and Over Identification Tests.

First-Stage Regression Summary Statistics					
Variable	R-sq	Adjusted R-sq	Partial R-sq	Cluster Robust F (2,5979)	Prob >F
MHHIdelta	0.9757	0.9755	0.085	25.6243	0
Hansen's J Over Identification Test					
Hansen's J	chi2(1) = 0.438032 (p = 0.5081)				

Note: This table shows the first-stage of of the two-stage least square estimation procedure used to implement the Instrumental Variable approach. Standard Errors are clustered at the state level.

2.3. Arellano-bond dynamic panel model

Arellano and Bond [35] provides a model to address endogeneity in dynamic panel models, using a General Methods of Moments (GMM) estimator. The ARB estimator is useful in situations where there are few time periods but many individual observations. The model is specified in equation 5;

$$\Delta log(Rate_{pjt}) = \beta_1 \Delta log(Rate_{pjt-1}) + \beta_2 \Delta MHHID_{jt} + \beta_3 \Delta HHI_{jt} + \beta_4 \Delta X_{jt} + \Delta \epsilon_{pjt} \quad (5)$$

Eq. 5 shows the ARB model. ARP model takes the logged dependent variable and uses lagged values of this variable as an instrument in the first stage to identify the endogenous regressors. Then the first difference is taken of that model to remove unobserved time invariant heterogeneity. Additionally, lagged values of MHHID are also added to the model's first stage as instruments to increase the explanatory power of the identified variable.

2.4. Bartik (shift-share) instruments

Bartik instruments, also referred to as shift share instruments, were

first proposed by Bartik [36] and have been refined over time in papers such as Jean and Katz [37], Adao et al. [38], and Goldsmith-Pinkham et al. [39]. Bartik instruments are usually used in economic papers that examine the effects of immigration on the economies of destination countries. These instruments are constructed by interacting the growth rate of migration inflows by country of origin with immigrants past migration inflows, taking into account geographic distribution [40].

I use the interaction of state level enrollment shares in ACA plans and national growth rates in enrollment. The Bartik instrument approach takes a measure at the aggregate level (the growth rate of national enrollment) and considers how variations in the degree to which different local shares (state level ACA plan enrollment) are exposed to this change given their relevant characteristics. The instrument is defined in equation 6;

$$B_{jt} = Z_{j0}G_t \tag{6}$$

B_{jt} is the Bartik instrument at time t and across states j and $Z_{j0}G_t$ represents the base year's local shares multiplied by the growth rate of national shares in year t. The Bartik instrument is then used in the first stage of the regression equation shown in equation 7;

$$MHHID_{jt} = \alpha_0 X_{jt} + \alpha_1 B_{jt} + \eta_{jt} \# \tag{7}$$

The ACA health insurance exchanges were opened in 2014. I use 2014 as the initial period, because insurance companies did not properly price their plans in 2014 due to the lack of market experience. I then shift the initial period by the growth rate of enrollments from 2015 to 2019. As a result, the Bartik instrument B_{jt} has an analogous structure to the difference-in-differences (DID) approach, where a base group is selected before an exogenous shock and that base group is compared against the groups post the shock occurring.

Results

Table 4 illustrates the results of the panel regression analysis, where I cluster corrected the standard errors at the state level. A one standard deviation increase in MHHI delta (MHHID) results in, on average, a 19 % increase in the monthly premiums paid for silver plans on the ACA exchanges. This is the result reported when using the Ordinary Least Squares (OLS) regression model. When using the FE approach, I find that the effect size of MHHI delta falls to 11 % after adding state and year fixed effects.

To give some context on the effect size of the OLS and FE estimators, the average monthly insurance premium paid for a silver plan is $500. So, a one standard deviation increase in MHHID leads to a $95 increase in the OLS model and a $55 increase in the FE model in the monthly premium. The effect size is in line with the results from other papers examining different industries. For example, Semov [29] finds that a one standard deviation increase in MHHID leads to an 8 % decrease in the cash holdings of commonly owned firms. In this study increases in CO caused a less competitive environment between commonly owned firms as these firms chose to hold fewer cash reserves because they were less concerned about a sudden negative liquidity event due to a decision by one of their competitors. Additionally, Azar et al. [6] finds that an increase in CO resulted in a 20 % increase in the price of airline tickets at the industry level, and a 4 % increase in prices at the route level.

[Insert Table 4 Here]

I also find that a one standard deviation increase in HHI increases premiums by 3 % in the OLS model and 2 % in the FE model. This result is in line with economic theory, where increases in the HHI lead to an increase in market concentration which decreases competition. Decreases in competition and increases in market power led to higher prices for consumers. Most of the other control variables are statistically insignificant. However, a one standard deviation increase in median household income is associated with a 0.4 % (OLS) and 18 % (FE) increase in insurance rates respectively. Higher income individuals receive fewer subsidies for health insurance and might choose more expensive plans. Variables for the type of insurance are also statistically significant. As expected, relative to the PPO plans, all other plans have lower premiums.

Table 4
The Effect of Common Ownership on Health Insurance Premiums Panel Regression.

Variables	(1) (OLS) log(Rate)	(2) (FE) log(Rate)
MHHI Delta HHI	0.190***	0.118*** (0.023) 0.017*
	(0.019)	(0.009)
	0.028***	
	(0.008)	
Enrollment in Original Medicare	-0.031	0.129
	(0.042)	(0.323)
Enrollment in Medicare Advantage	0.010	0.168
	(0.043)	(0.137)
Bachelors or Higher	0.050	0.022
	(0.046)	(0.046)
Some College	-0.208**	-0.112
	(0.092)	(0.111)
High School Graduate	0.194**	0.077
Median Household Income	(0.083) 0.004*	(0.118) 0.180*
	(0.029)	(0.093)
Poverty Rate for Children Under 5	0.061	0.011
	(0.044)	(0.048)
Poverty Rate	-0.093*	-0.057
	(0.054)	(0.087)
GDP per-capita	-0.005	0.038
	(0.018)	(0.109)
Mean Out of Pocket Spending	-0.004	-0.009
Wellness Program Offered	(0.007)	(0.016) 0.013**
	-0.017** (0.007)	(0.006)
Obesity Rate	-0.009	0.008
EPO HMO	(0.015)	(0.021)
	-0.135***	-0.172*** (0.022)
	(0.034)	-0.096*
	-0.105***	(0.0542)
	(21.10)	
POS	-0.034	-0.078
Constant	(0.030)	(0.054)
	6.257***	6.223*** (0.035)
	(0.021)	
Observations	6,039	6,039
R-squared	0.704	0.738
State FE	No	Yes
Year FE	No	Yes

Cluster corrected standard errors at the state level are in parentheses
*** p < 0.01, ** p < 0.05, * p < 0.1

Table 5
OLS, FE, IV, and ARB: Panel Estimation Results.

VARIABLES	(OLS) log(Rate)	(FE) log(Rate)	(IV) log(Rate)	(ARB) log(Rate)
MHHI Delta	0.190***	0.118***	0.107**	0.077**
HHI	(0.019)	(0.023) 0.017*	(0.054)	(0.037)
	0.028***	(0.009)	0.036***	0.030***
	(0.008)		(0.004)	(0.009)
Observations	6,039	6,039	6,037	5,741
R-squared	0.6999	0.7381	0.5927	0.5970
State FE	No	Yes	Yes	N/A
Year FE	No	Yes	Yes	N/A

Cluster corrected standard errors at the state level are in parentheses
*** p < 0.01, ** p < 0.05, * p < 0.1

3.1. Instrumental variable approach, and Arellano-bond dynamic panel model estimation

[Insert Table 5 Here]

Table 5 shows the comparison between the OLS, FE, IV, and ARB models. For the IV model, I use lags of MHHID as instruments. In the ARB panel regression model, which is in first differences I use a lag of the dependent variable and two lags of the CO variable as instruments. The results from the IV model show that increasing MHHID by one standard deviation leads to an increase in premium rate of 10.7 % on average. This estimate is similar to the estimate from the FE model but is slightly smaller. I also find that the impact of HHI is larger in the IV model. Finally, the estimate from the ARB model indicates that a one standard deviation increase in MHHID leads to a 7.7 % increase in monthly insurance rates, on average.

3.2. Bartik instrument estimation

[Insert Table 6 Here]

Table 6 contains estimates from the Bartik instrument model. A one standard deviation increase in MHHID results in a 16.7 % increase in insurance premiums. This estimate is closer in magnitude to the OLS estimates than the FE, IV, and ARB models. There appears to be an upward bias in the OLS and Bartik instrument estimates. The OLS estimates upward bias could be due to many potentially omitted variables and the endogeneity of the CO measure. The Bartik instrument approach could have upwardly biased estimates due to selection in local shares, as some states may provide more support for enrolling in ACA plans, making plan enrollment endogenous. Irrespective of the individual model, estimates from the OLS, FE, IV, ARB, and Bartik instrument models all indicate that increases in MHHID lead to an increase in insurance premiums. This result provides evidence that increases in CO lead to anti-competitive behavior.

3.3. Alternative measures of Co

To test the robustness of the results, I use two alternative measures of CO. The first alternative measure uses the density of CO (Density) created by taking the ratio of the number of firm-pairs that are connected in an industry over the maximum possible number of firm-pairs in that industry [41]. Firm-pairs are connected when there is an institutional investor that owns at least 5 % of each of the two firms. The density measure captures the intensity of CO between firms in the industry.

The second alternative measure of CO I examine is the Percentage of Common Funds (PCF), defined as the number of common owners who own at least 5 % of two or more firms in an industry divided by the total number of common owners in the industry [41]. If no institutional investor owns two or more companies with at least 5 % ownership, then PCF will be zero. PCF captures the behavior of institutional investors that have significant stakes in multiple firms in the same industry.

Table 6
Estimation results: Bartik Instrument.

Variable	Coefficient
(Std. Err.)	
MHHI Delta	0.167**
(0.032)	
HHI	0.017
(0.011)	
Observations	5,140
R-squared	0.78
State FE	Yes
Year FE	Yes
Cluster corrected standard errors are in parentheses	

*** p < 0.01, ** p < 0.05, * p < 0.1

Table A1 shows estimates using the two alternatives' measures of CO. The magnitude of PCF estimates is similar to MHHID. However, estimates from Density are lower in magnitude and less precise, owing in part to more limited variation over time. There is only a hand full of institutional blockholders that exist in any industry.

Table A2 shows another alternative measure of CO using a firm level CO measure taken from He and Huang [25]. Cross Ownership is a binary variable that equals one if the firm is cross held in any of the four quarters in a sample year and zero otherwise. The results show that Cross Ownership has a statistically insignificant effect on insurance rates in all models. In most quarters, firms in the sample were cross held, so this measure had limited variation over the sample period.

Discussion

The effect of Modified Herfindahl-Hirschman Index Delta (MHHID) on insurance premiums from my models is both statistically and economically significant. An increase in MHHID leads to an increase in insurance premiums. Combining all the results from Tables 5 and 6 brings the range of point estimates from 7.7 % to 19 %. The estimates from the Bartik, FE, IV, and ARB models indicate that the OLS coefficient of MHHID is overestimated. This is likely because investors are more likely to buy shares of companies with greater pricing power ([6]a).

The results from the estimation strategies used in this paper are in line with other identified results from the CO literature. For instance, Azar et al. [6] found using a DID framework that created a treatment group containing airlines that were jointly owned by Blackrock and BGI before the acquisition and a control group containing all commonly owned airlines. The DID model showed that airline ticket prices were 10 % to 12 % higher, compared against a world in which firms were not commonly owned or one in which managers ignore the anti-competitive incentives of common owners.

What is clear from the empirical results is that policymakers like the SEC should take the anti-competitive effects of CO more seriously. It is possible that the increase in product prices and decreases in competition could incentivize other institutional investors to enter more industries and establish themselves as common owners. For instance, a recent Wall Street Journal article showed that institutional investors were purchasing large stakes in life insurance companies in order to have a significant number of shares in the biggest firms in that industry.

With respect to what policymakers and regulators should do, it seems clear that more data needs to be made available to researchers. As stated previously, most of the data that has been used in the literature comes from the SECs 13F filings, which are known to have significant quality issues, and only contain firms with at least $100 million dollars in assets under management and are only collected quarterly. The SEC should change its rules on filing share ownership data so that it is more frequent, comprehensive, and covers firms that have fewer assets. This regulatory change would go a long way to increasing the quantity and quality of CO research.

Finally, what is left unexplored until now are the mechanisms that common owners use to encourage this anti-competitive behavior. Even though many common owners specialize in passive investments, they are not truly "passive" in key areas [42]. For example, they are not passive in corporate governance, which is the is most common direct channel that has been found in the literature [43]. Large institutional investors vote in shareholder meetings on key issues like executive compensation and corporate strategies (Becht, Franks and Wagner, 2023). This is the case in the health insurance industry, where institutional investors like Vanguard and Blackrock vote on behalf of their funds shareholders. Additionally, institutional investors have dedicated teams that develop proxy voting guidelines, which can shape how firms and their managers respond to governance and competitive strategy issues [44]. Fichtner, Heemskerk, and Garcia [45] found that because of these voting guidelines, executives at these firms are prone to internalize the objectives of the largest investors like Blackrock, Vanguard, and

State Street.

Another channel that common owners use is that because they have a collective interest in the firms in their diversified portfolios, they will pursue portfolio value maximization strategies instead of firm value maximization strategies [43]. For instance, common owners will advocate for greater disclosure for firms in their portfolios [24]. With more information competitors can better coordinate their responses to actions taken by the revealing firm. This would increase industry coordination and could reduce competition, benefiting common owners. A further analysis of the mechanisms used by common owners in health insurance is an important area of my future research.

Conclusion

To conclude, I find using the Arellano-Bond Dynamic panel model and Bartik instruments that an increase in common institutional ownership leads to higher premiums paid by policyholders for their silver plans on the ACA exchanges. This result is robust to the use of alternative measures of common ownership. The results are consistent with theory suggesting that common owners influence the decisions that managers in commonly owned firms make, incentivizing anti-competitive behavior.

More research needs to be conducted to examine the mechanisms and channels through which common owners promote anti-competitive behavior. Policymakers should address the ability of institutional investors to become common owners or enact polices that mitigate the anti-competitive effects of common institutional ownership.

CRediT authorship contribution statement

Kevin Smith: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.healthpol.2025.105316.

References

[1] Wager Emma, Ortaliza Jared, Cox Cynthia. How does health spending in the U.S. compare to other countries? Peterson-KFF Health System Tracker 2022. Feb.
[2] Woolf Steven H, Schoomaker Heidi. Life expectancy and mortality rates in the United States, 1959-2017. JAMA 2019;322(20):1996–2016.
[3] Cutler David M, Ly Dan P. The (paper) work of medicine: understanding international medical costs. J Econ Perspect 2011;25(2):3–25.
[4] Glied Sherry A, Altman Stuart H. Beyond antitrust: health care and health insurance market trends and the future of competition. Health Aff 2017;36(9):1572–7.
[5] Newitt Patsy. 5 largest health insurance companies by membership. Becker's ASC Rev 2021. Mar.
[6] Azar José, Schmalz Martin C, Tecu Isabel. Anticompetitive effects of common ownership. J Finance 2018;73(4):1513–65.
[7] O'Brien Daniel P, Salop Steven C. Competitive effects of partial ownership: financial interest and corporate control. Antitrust LJ 1999;67:559.
[8] Nasdaq. Nasdaq; Institutional Stock Holdings. Nasdaq 2024.
[9] Backus, Matthew, Christopher T Conlon, and Michael Sinkinson, "Common Ownership data: scraped SEC form 13F filings for 1999-2017," 2020.
[10] Lewellen Katharina, Lowry Michelle. Does common ownership really increase firm coordination? SSRN 2019:3336343.
[11] Frean Molly, Gruber Jonathan, Sommers Benjamin D. Premium subsidies, the mandate, and Medicaid expansion: coverage effects of the Affordable Care Act. J Health Econ 2017;53:72–86.
[12] Parys Jessica Van. ACA marketplace premiums grew more rapidly in areas with monopoly insurers than in areas with more competition. Health Aff 2018;37(8):1243–51.
[13] Gruber, Jonathan and Benjamin D Sommers, "The Affordable Care Act's Effects on Patients, Providers and the Economy: What We've Learned So Far," Technical Report, National Bureau of Economic Research 2019.
[14] Levitt Larry, Claxton Gary, Damico Anthony, Cox Cynthia. Assessing ACA marketplace enrollment. Kaiser Family Foundation 2016.
[15] Dafny Leemore, Gruber Jonathan, Ody Christopher. More insurers lower premiums: evidence from initial pricing in the health insurance marketplaces. Am J Health Econ 2015;1(1):53–81.
[16] Mcdermott Daniel, Cox Cynthia. Insurer participation on the ACA Marketplaces, 2014- 2021. Kaiser Family Foundation 2021.
[17] Jacobs Paul D, Hill Steven C. ACA marketplaces became less affordable over time for many middle-class families, especially the near-elderly: study examines ACA marketplaces affordability. Health Aff 2021;40(11):1713–21.
[18] Scheffler Richard M, Arnold Daniel R, Fulton Brent D, Glied Sherry A. Differing impacts of market concentration on Affordable Care Act Marketplace premiums. Health Aff 2016;35(5):880–8.
[19] Hiltzik Michael. U.S. judge finds that Aetna deceived the public about its reasons for quitting Obamacare. Los Angeles Times 2017. Jan.
[20] Rotemberg, Julio, "Financial transaction costs and industrial performance," 1984.
[21] Dennis Patrick J, Gerardi Kristopher, Schenone Carola. Common ownership does not have anti-competitive effects in the airline industry. *Available at* SSRN 2019;3063465.
[22] Kennedy, Pauline, Daniel P O'Brien, Minjae Song, and Keith Waehrer, "The competitive effects of common ownership: economic foundations and empirical evidence," 2017.
[23] Azar, José, Martin C Schmalz, and Isabel Tecu,"Reply to: 'common ownership does not have anti-competitive effects in the airline industry'," *Available at SSRN 3168095*, 2018.
[24] Schmalz Martin C. Recent studies on common ownership, firm behavior, and market outcomes. Antitrust Bullet 2021;66(1):12–38.
[25] He Jie Jack, Huang Jiekun. Product market competition in a world of cross-ownership: evidence from institutional blockholdings. Rev Financ Stud 2017;30(8):2674–718.
[26] Irani, Mohammad, Vahid, Yang, Wenhao, Zhang, and Feng, "Common ownership and competition in mergers and acquisitions," Oct 2019.
[27] Antón Miguel, Ederer Florian, Giné Mireia, Schmalz Martin. Innovation: the bright side of common ownership? Manage Sci 2024.
[28] Abraham Jean, Drake Coleman, WSacks Daniel, Kosali Simon. Demand for health insurance marketplace plans was highly elastic in 2014–2015. Econ Lett 2017;159:69–73.
[29] Semov Svetoslav. Common ownership, competition and firm financial policy. Competit Firm Financial Policy 2017. *April 20, 2017*.
[30] Diaz, Maria, Jennifer Tolbert, and Cornelia Hall, "State actions to improve the affordability of health insurance in the individual market," KFF, Jul 2019.
[31] Kwon, Heung Jin, "Executive compensation under common ownership.," 2017.
[32] Connelly Brian L, Lee Kang Bok, Tihanyi Laszlo, Certo S Trevis, Johnson Jonathan L. Something in common: competitive dissimilarity and performance of rivals with common shareholders. Acad Manage J 2019;62(1):1–21.
[33] Tihanyi Laszlo, AJohnson Richard, Hoskisson Robert E, Hitt Michael A. Institutional ownership differences and international diversification: the effects of boards of directors and technological opportunity. Acad Manage J 2003;46(2):195–211.
[34] Stock, James H and Motohiro Yogo, "Testing for weak instruments in linear IV regression," 2002.
[35] Arellano Manuel, Bond Stephen. Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. Rev Econ Stud 1991;58(2):277–97.
[36] Bartik, Timothy J, "Who benefits from state and local economic development policies?" 1991.
[37] Jean Blanchard Olivier, Katz Lawrence F. "Regional evolutions,". Brookings Papers Econom Activity, Econom Studies Program, Brookings Institut 1992;23(1).
[38] Adao Rodrigo, Kolesár Michal, Morales Eduardo. Shift-share designs: theory and inference. Q J Econ 2019;134(4):1949–2010.
[39] Goldsmith-Pinkham Paul, Sorkin Isaac, Swift Henry. Bartik instruments: what, when, why, and how. Am Econom Rev 2020;110(8):2586–624.
[40] Jaeger, David A, Joakim Ruist, and Jan Stuhler, "Shift-share instruments and the impact of immigration," Technical Report, National Bureau of Economic Research 2018.
[41] Koch Andrew, Panayides Marios, Thomas Shawn. Common ownership and competition in product markets. J financ econ 2021;139(1):109–37.
[42] Partnoy Frank. Are Index funds evil? Atlantic 2017.
[43] Tzanaki Anna. Varieties and mechanisms of common ownership: A calibration exercise for competition policy. J Compet Law Econ 2022;18(1):168–254.
[44] Kerber Ross. Vanguard to double investor 'voting choice' Program for 2025. Reuters 2024.
[45] Fichtner Jan, Heemskerk Eelke M, Garcia-Bernardo Javier. Hidden power of the Big three? Passive index funds, re-concentration of corporate ownership, and new financial risk. Bus Polit 2017;19:298–326.